

Autonomous Vehicle Lane Following (EKF): Fusion of IMU and Camera/GPS Sensors

Multi-rate sensor fusion with bias tracking, NIS gating and Stanley control

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Presentation Roadmap / Required Coverage

- ① Clear project motivation
- ② Process model
- ③ Measurement model
- ④ Filter design/details
- ⑤ Comparative results
- ⑥ Demonstration of the system

Project Motivation: Why Sensor Fusion?

Lane following is safety-critical

- Estimate position, velocity and heading continuously.
- A single sensor is not reliable under all road conditions.
- The controller uses estimated states, not ground truth.

Sensor limitations

- **IMU:** high-rate, but drifts due to bias.
- **GPS:** absolute position, but low-rate and can disappear in tunnels.
- **Camera:** lane-relative information, but frame drops and glare outliers occur.

Goal: robust lane following by fusing IMU + GPS + camera in an EKF.

Multi-rate estimation pipeline

- IMU prediction: **100 Hz**
- GPS correction: **10 Hz**
- Camera correction: **30 Hz**
- Estimated states feed a closed-loop Stanley controller.

6-DOF state vector

$$\mathbf{x}_k = [p_x, p_y, v, \psi, b_\omega, b_a]^T$$

Includes vehicle motion and slow-varying IMU hardware biases.

Discrete-time motion model

$$p_{x,k} = p_{x,k-1} + v_{k-1} \cos \psi_{k-1} \Delta t$$

$$p_{y,k} = p_{y,k-1} + v_{k-1} \sin \psi_{k-1} \Delta t$$

$$v_k = v_{k-1} + (a_m - b_a) \Delta t$$

$$\psi_k = \psi_{k-1} + (\omega_m - b_\omega) \Delta t$$

Bias random walk

$$b_{\omega,k} = b_{\omega,k-1} + w_\omega, \quad b_{a,k} = b_{a,k-1} + w_a$$

- Base step: $\Delta t = 0.01$ s.
- Bias states enable dead reckoning during GPS outage.
- Analytic Jacobian $F = \partial f / \partial x$ propagates covariance.

Measurement Models: GPS and Camera Updates

GPS absolute position

$$z_{gps} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \end{bmatrix} x + v_{gps}$$

- Directly observes p_x, p_y .
- Noise: $\sigma_{gps} = 2.0$ m.
- Includes tunnel dropout and multipath artifacts.

Camera lane-relative observation

$$CTE = -\sin(\psi_{lane})dx + \cos(\psi_{lane})dy$$

$$h_{rel} = \psi - \psi_{lane}$$

- Measures cross-track error and relative heading.
- Noise: $\sigma_{CTE} = 0.20$ m, $\sigma_h = 0.06$ rad.
- Includes frame drops and glare outliers.

Why EKF?

- Nonlinear bicycle model, but analytically tractable.
- Lower computational load than PF/UKF for 100 Hz IMU.
- Suitable for deterministic real-time estimation.

Joseph covariance update

$$P = (I - KH)P(I - KH)^T + KRK^T$$

- Maintains symmetry and positive definiteness.
- Reduces covariance collapse/divergence risk.

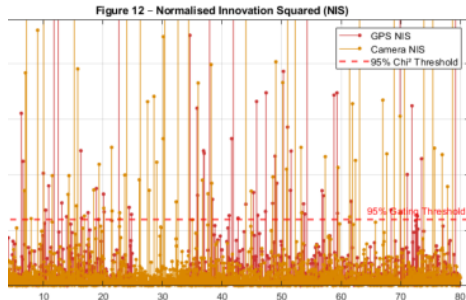
Asynchronous logic

Predict every IMU step. Correct only when GPS or camera measurements arrive.

Normalized Innovation Squared

$$\epsilon_z = v_k^T S_k^{-1} v_k$$

- Innovation is tested statistically.
- If $\epsilon_z > 5.991$ for 2 DOF at 95%, the measurement is rejected.
- Protects EKF from GPS multipath and camera glare spikes.



NIS spikes crossing the threshold are rejected.

Closed-Loop Control and Demonstration Plan

Stanley controller

$$\delta_k = e_{\psi,EKF} + \arctan\left(\frac{ke_{CTE,EKF}}{v_{EKF}}\right)$$

- Uses EKF estimates only.
- Velocity clamp prevents singularity near $v = 0$.
- Steering saturated at $\pm 35^\circ$.

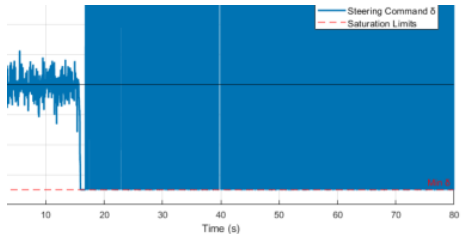


Fig. 10. Controller steering commands computed from EKF estimates

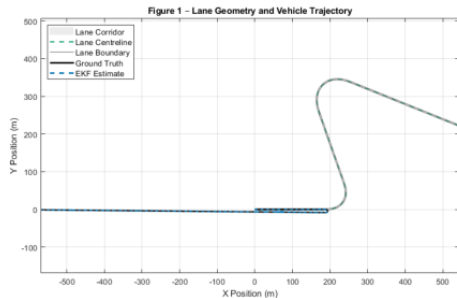
Steering command remains inside mechanical limits.

Demo clip to show in video

Show the simulation: lane geometry, EKF trajectory, GPS tunnel dropout, and controller response.

Comparative Results I: Trajectory and Tunnel Dropout

resilience.



Overall lane geometry and EKF trajectory.



Fig. 4. Zoomed view of the GPS tunnel dropout region. The EKF flawlessly executes dead reckoning, maintaining trajectory stability until GPS is restored.

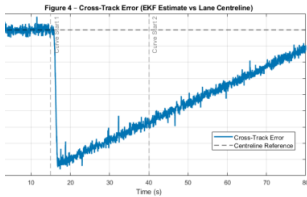
B. Lateral and Longitudinal Tracking Analysis

The closed-loop tracking performance generated by EKF-fed Stanley Controller is highly stable. The Cross-Track Error (Fig. 5) shows that the vehicle successfully mitigates curve-induced overshoots and rarely deviates more than a centimeter from the centerline target. Similarly, limited

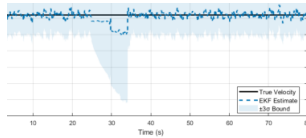
Zoomed GPS tunnel dropout region.

- EKF continues with IMU dead reckoning when GPS is unavailable.
- Camera CTE helps maintain lateral lane position.
- Trajectory recovers when GPS returns.

Comparative Results II: Tracking and Bias Estimation

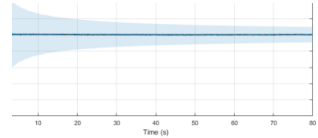


Cross-track error



Velocity tracking (True vs EKF). The 100 Hz kinematic update ensures velocity estimates tightly bounded by $\pm 3\sigma$ covariance limits.

Velocity tracking



Gyroscope Bias Estimation. The filter reliably tracks the random walk and remains securely within the estimated covariance envelope.

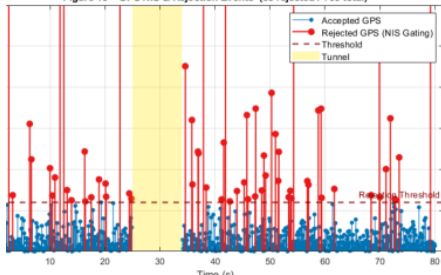
Gyroscope bias

Interpretation

The controller maintains lane following, while EKF bias estimation prevents heading and velocity drift during measurement interruptions.

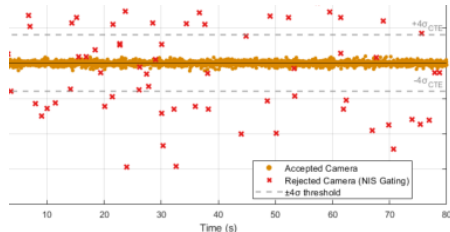
Robust Outlier Rejection Results

Figure 15 – GPS NIS & Rejection Events (63 rejected / 709 total)



GPS multipath/outlier rejection.

Result: corrupted measurements are detected before they pull the estimate away from the lane.



Camera glare residual rejection.

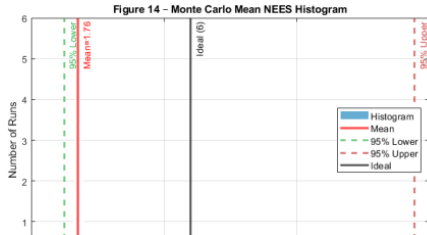
Monte Carlo and Filter Consistency

12. Monte Carlo RMSE Histograms across 50 independent runs, illustrating robust error bounding under varying stochastic noise profiles.

VI. CONCLUSION

This study successfully developed and mathematically validated an Extended Kalman Filter for the highly asynchronous

RMSE histograms over 50 Monte Carlo runs.



Mean NEES near the 6-DOF ideal.

Consistency check

NEES evaluates whether EKF uncertainty is realistic. For this 6-state filter, the expected average value is close to 6.

Conclusion

- A 6-DOF EKF was designed for lane following using IMU, GPS and camera measurements.
- Bias estimation enables stable dead reckoning during GPS tunnel dropout.
- Joseph covariance update improves numerical stability.
- NIS gating rejects GPS and camera outliers before they corrupt the state.
- EKF estimates successfully drive the Stanley controller in closed loop.

Submission reminder

Upload the video and confirm the link is public or shared with the instructor before submitting.

- [1] M. Arulampalam et al., “A tutorial on particle filters for online nonlinear/non-Gaussian Bayesian tracking,” IEEE TSP, 2002.
- [2] S. Julier and J. Uhlmann, “Unscented filtering and nonlinear estimation,” Proceedings of the IEEE, 2004.
- [3] K. Jo, K. Chu, and M. Sunwoo, “IMM filter-based sensor fusion of GPS with in-vehicle sensors,” IEEE T-ITS, 2012.
- [4] S. H. Jo et al., “Localization of a car based on multi-sensor fusion,” ISPRS Archives, 2018.
- [5] C. Blanco et al., “Autonomous ground vehicle lane-keeping LPV model-based control,” Sensors, 2021.

Thank You

Questions?

Developed for the Kalman and Bayesian Filters course.